MINOR IN ARTIFICIAL INTELLIGENCE

Rationale

The College of Engineering/ECE Department offers a minor in Artificial Intelligence (AI). The minor aims to meet the anticipated national, regional, and international demand for experts in Artificial Intelligence, Knowledge Representation, Data Mining, Machine Learning, and Automated Reasoning.

Minor Objectives

The aims of the minor are to:
1. Provide students with advanced expertise in AI.
2. Prepare students to apply AI in industry.
3. Prepare students for graduate studies in AI.

Learning Outcomes

At the end of this minor, the student is expected to demonstrate the ability to:
1. Use current techniques, architectures, and tools to build intelligent systems.
2. Design systems and tools to evaluate the performance of intelligent systems.
3. Apply AI tools and techniques to improve the performance of existing non-AI-enabled systems.

Minor Requirements

While most relevant to students in Computer and Communications Engineering and Computer Science, this minor is also offered to undergraduate RHU students in the BE or MS program from other disciplines (Engineering and Computer Science). It may also be offered to students with a BS/BE from other universities, subject to a case-by-case evaluation of their transcripts and other specific RHU requirements.

Early in their major, interested RHU engineering students need to complete the appropriate form declaring their intent to minor in AI while completing their regular major.

To complete the AI minor, a student must:
1. Declare a minor in Artificial Intelligence by completing the Minor Declaration Form.
2. Obtain the approval of the dean of the student's college and the dean of the College of Engineering.
3. Earn a Minor Cumulative Grade Point Average of no less than 70 %.
4. Complete 18 credits of coursework as specified below.

It should be noted that overlap among student major requirements, other minor requirements, and AI minor requirements is counted toward fulfilling the minor

requirements. The student must complete a minimum of 3 credit hours of coursework that is not counted toward the requirement for their major or any other minor.

Career Options

This minor allows its holders to pursue careers in a variety of sectors, regardless of their major. These sectors include, but are not limited to, communications, computers, computer networks, environmental engineering, energy, smart cities, autonomous vehicles, biomedical engineering, health, security, biology, and physics.

Curriculum and Program

The Artificial Intelligence minor consists of six courses (18 credits), two of which are mandatory and four electives selected to satisfy the requirements of the proposed program objectives and learning outcomes.

Mandatory Courses

The three mandatory courses (9 credits) are:

CCEE 460	Artificial Intelligence	3(3,0)
CCEE 564	Machine Learning and Data Mining	3(3,0)

Elective Courses

Students may choose three elective courses (9 credits) from the following list.

CCEE 516	Advanced Programming and Database Management Systems	3(2,2)
CCEE 536	Digital Image Processing	3(3,0)
CCEE 561	Computer Vision	3(3,0)
CCEE 566	Natural Language Processing	3(3,0)
CCEE 567	Optimization	3(3,0)
CCEE 568	Big Data and Analytics/Big Data Systems	3(3,0)
CCEE 612	Advanced Data Mining	3(3,0)

Courses Description

CCEE 460	**Artificial Intelligence**	**3(3,0)**
This course introduces students to the fundamentals of knowledge representation and learning in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include intelligent agents, informed and uninformed search strategies, and adversarial search.		

The Python language libraries will also be introduced. **Prerequisite**: CCEE 214 or equivalent.

CCEE 516	**Advanced Programming and Database Management Systems**	**3(2,2)**
This course will introduce students to database management and querying, as well as to data warehousing. Students will also learn advanced programming tools, including Bigtable, NoSQL, R, Python, Scala, MapReduce, and Elasticsearch, and apply them to address big data challenges. **Prerequisite**: CCEE 315 or equivalent.		

CCEE 536	**Digital Image Processing**	**3(2,2)**
Two-dimensional signals and systems. Image formation and perception. Representation, coding, filtering, restoration, and enhancements. Feature extraction and scene analysis. Introduction to computer vision. Introduction to Medical Imaging. **Prerequisite**: CCEE 214		

CCEE 561	**Computer Vision**	**3(3,0)**
This course introduces the principles, models, and applications of computer vision. The course will cover image structure, projection, stereo vision, and the interpretation of visual motion. Case studies of industrial (robotic) applications of computer vision, including visual navigation for autonomous robots, robot hand-eye coordination, and novel man-machine interfaces. **Prerequisite**: CCEE 214 or COSC 214.		

CCEE 563	**Robotics**	**3(3,0)**
Introductory historical development of robotics; rigid-object kinematics; robot arm kinematics; inverse kinematics; dynamics; introduction to trajectory planning and manipulator control. **Equivalent to**: MECA 544. **Prerequisite**: None		

CCEE 564	**Machine Learning and Data Mining**	**3(3,0)**
This course introduces students to the basic knowledge representation and learning techniques. The emphasis is on understanding the data mining process and applying the corresponding approaches to solve practical problems and develop intelligent software applications. The course covers several topics related to classification, prediction, and clustering. **Prerequisite:** CCEE 214		

CCEE 566	**Natural Language Processing**	**3(3,0)**
This course introduces students to the field of natural language processing (NLP). The student is first introduced to word and sentence tokenization. The student then uses the learned skills to implement systems for text classification and sentiment analysis, spelling correction, information extraction, parsing, meaning extraction, and question answering, Machine learning algorithms as well as algorithms like n-gram language modeling, naive bayes and maxent classifiers, sequence models like Hidden Markov Models, probabilistic dependency and constituent parsing, and vector-space models of meaning will be introduced as needed for the above NLP applications. **Prerequisite**: CCEE 214		

CCEE 567	Optimization	3(3,0)
This course introduces students to the theory, algorithms, and applications of optimization. The optimization methodologies include linear programming, network optimization, integer programming, and decision trees. It includes a team project in which students select and solve a problem in practice. **Equivalent to**: BADM 420 and MECH 571. **Prerequisite**: MATH 215 or equivalent.		

CCEE 568	Big Data and Analytics/Big Data System	3(3,0)
This course enables students to understand why the Big Data Era has emerged. Students will become conversant in the terminology and core concepts of big data problems, applications, and systems. Students will learn how to make Big Data useful in their business or career. Students will be introduced to one of the most common frameworks, Hadoop, which has made big data analysis easier and more accessible -- increasing the potential for data to transform our world. **Prerequisite**: CCEE 564 or equivalent.		